

CPE 334 INTRODUCTION TO SOFTWARE ENGINEERING IN THE AGE OF AI AGENTS

Course: CPE 334 (Semester 1/2026)

Lab Increment: Lab 3 (60 Points)

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(@Suprawi5227)

Peer Repository: Suprawi5227/toktickit

Submission Date: October 4, 2026

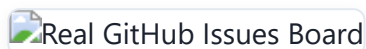
Answer Part 1: Git Use with Engineering Workflow

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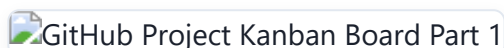
The TokTickIT Lab 3 engineering increment extends the single-context requester prototype into a production-grade enterprise IT Service Desk with multi-role authentication (JWT, bcrypt, mandatory password reset), role-based access control, IT Staff Ticket Queue and Internal Notes, and an Administrator User Management Console with safety rules.

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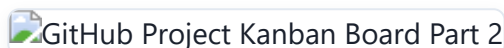
All 10 feature issues for Lab 3 (Issue #34 to Issue #52) were managed using a 10-Issue staged integration workflow. Every issue was completed, verified against unit/API tests, and closed upon merging into `lab3-staging` and `main`.



Real GitHub Issues Board



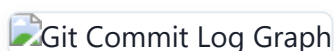
GitHub Project Kanban Board Part 1



GitHub Project Kanban Board Part 2

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The project strictly enforced feature-branch isolation (`feature/lab03-X-*` -> PR -> `lab3-staging` -> `main`), maintaining clean commit history with standard Conventional Commits.



Git Commit Log Graph

Table 1: Pull Requests Given (natthakamol1130/toktickit)

Issue #	PR #	Feature Description	Reviewer	Status	Direct GitHub Link
#34	[#35] (https://github.com/natthakamol1130/toktickit/pull/35)	Lab 3 Specification, UI Spec, and API Spec Documents	@Suprawi5227	Merged	[PR #35] (https://github.com/natthakamol1130/toktickit/pull/35)
#36	[#37] (https://github.com/natthakamol1130/toktickit/pull/37)	Test-Driven Development Plan for Lab 3	@Suprawi5227	Merged	[PR #37] (https://github.com/natthakamol1130/toktickit/pull/37)
#38	[#39] (https://github.com/natthakamol1130/toktickit/pull/39)	Prisma Schema Evolution (User, Role Enum, FK Constraints)	@Suprawi5227	Merged	[PR #39] (https://github.com/natthakamol1130/toktickit/pull/39)
#40	[#41] (https://github.com/natthakamol1130/toktickit/pull/41)	Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	@Suprawi5227	Merged	[PR #41] (https://github.com/natthakamol1130/toktickit/pull/41)
#42	[#43] (https://github.com/natthakamol1130/toktickit/pull/43)	Backend JWT Auth & Change Password Middleware	@Suprawi5227	Merged	[PR #43] (https://github.com/natthakamol1130/toktickit/pull/43)
#44	[#45] (https://github.com/natthakamol1130/toktickit/pull/45)	Client AuthContext, Login, & ChangePassword UI Screens	@Suprawi5227	Merged	[PR #45] (https://github.com/natthakamol1130/toktickit/pull/45)

#46	[#47] (https://github.com/natthakamol1130/toktickit/pull/47)	Requester Ticket & Comment Access Control APIs	@Suprawi5227	Merged	[PR #4 (https://github.com/natthakamol1130/toktickit/pull/47)
#48	[#49] (https://github.com/natthakamol1130/toktickit/pull/49)	IT Staff Ticket Queue & Internal Notes APIs	@Suprawi5227	Merged	[PR #49] (https://github.com/natthakamol1130/toktickit/pull/49)
#50	[#51] (https://github.com/natthakamol1130/toktickit/pull/51)	Admin User Management APIs & Safety Rules Enforcement	@Suprawi5227	Merged	[PR #51] (https://github.com/natthakamol1130/toktickit/pull/51)
#52	[#53] (https://github.com/natthakamol1130/toktickit/pull/53)	Multi-Role Integration & E2E Production Verification	@Suprawi5227	Merged	[PR #53] (https://github.com/natthakamol1130/toktickit/pull/53)
Release	[#55] (https://github.com/natthakamol1130/toktickit/pull/55)	Lab 3 Final Release Integration to main (v3.0.0)	@Suprawi5227	Open/Ready	

Table 2: Pull Requests Reviewed (Suprawi5227/toktickit)

Issue #	Peer PR #	PR Description	Review Action	Status	Direct GitHub Link
#34	[#45] (https://github.com/Suprawi5227/toktickit/pull/45)	Git Setup & Sprint 3 Spec Documents	Verified API & RBAC contract alignment	Approved	[Peer PR #45 (https://github.com/Suprawi5227/toktickit/pull/45)
#36	[#47] (https://github.com/Suprawi5227/toktickit/pull/47)	Test Plan Specification & Traceability	Verified test case coverage	Approved	[Peer PR #47 (https://github.com/Suprawi5227/toktickit/pull/47)

#38	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Database Schema Evolution & User Models	Verified User model & FK constraints	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#40	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Idempotent Seed Data & Bcrypt Hashing	Verified bcrypt salt rounds & idempotent seeds	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#42	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Backend Auth APIs & Session Middleware	Verified JWT token validation & 401 handling	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#44	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Client Auth Context & Login UI	Verified form state management & logout cleanup	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#46	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	Requester Ticket Access Control APIs	Verified 403 authorization boundary	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#48	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	IT Staff Ticket Queue & Internal Notes APIs	Verified internal notes leak protection	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#50	[#52] (https://github.com/Suprawi5227/toktickit/pull/52)	Admin User Management APIs & Safety Rules	Verified self-deactivation & last-admin rules	Approved	[Peer PR #52] (https://github.com/Suprawi5227/toktickit/pull/52)

#52	[#53] (https://github.com/Suprawi5227/toktickit/pull/53)	Zen Green UI Theme Consistency & Signoff	Verified Zen Green UI & E2E integration	Approved	[Peer PR #5 (https://gitl)
Release	[#54] (https://github.com/Suprawi5227/toktickit/pull/54)	Sprint 3 Final Release Integration	Verified full Sprint 3 integration to main	Approved	[Peer PR (https:///)

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Issue 1: Lab 3 Specification DD Documents (Issue #34)

PR Received (natthakamol1130/toktickit/pull/35)



Real PR Received Issue 1

PR Given (Suprawi5227/toktickit/pull/45)



Real PR Given Issue 1

Issue 2: Lab 3 Test-Driven Development Plan (Issue #36)

PR Received (natthakamol1130/toktickit/pull/37)



Real PR Received Issue 2

PR Given (Suprawi5227/toktickit/pull/47)



Real PR Given Issue 2

Issue 3: Database Schema Evolution (Issue #38)

PR Received (natthakamol1130/toktickit/pull/39)



Real PR Received Issue 3

PR Given (Suprawi5227/toktickit/pull/49)



Real PR Given Issue 3

Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue #40)

PR Received (natthakamol1130/toktickit/pull/41)



Real PR Received Issue 4

PR Given (Suprawi5227/toktickit/pull/49)



Real PR Given Issue 4

Issue 5: Backend Auth & Change Password Middleware APIs (Issue #42)

PR Received (natthakamol1130/toktickit/pull/43)



Real PR Received Issue 5

PR Given (Suprawi5227/toktickit/pull/49)



Real PR Given Issue 5

Issue 6: Client Auth UI & Context Integration (Issue #44)

PR Received (natthakamol1130/toktickit/pull/45)



Real PR Received Issue 6

PR Given (Suprawi5227/toktickit/pull/49)



Real PR Given Issue 6

Issue 7: Requester Ticket & Comment Access Control APIs (Issue #46)

PR Received (natthakamol1130/toktickit/pull/47)



Real PR Received Issue 7

PR Given (Suprawi5227/toktickit/pull/51)



Real PR Given Issue 7

Issue 8: IT Staff Ticket Queue & Internal Notes APIs (Issue #48)

PR Received (natthakamol1130/toktickit/pull/49)



Real PR Received Issue 8

PR Given (Suprawi5227/toktickit/pull/51)



Real PR Given Issue 8

Issue 9: Admin User Management APIs & Safety Rules (Issue #50)

PR Received (natthakamol1130/toktickit/pull/51)



Real PR Received Issue 9

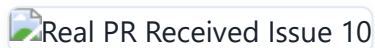
PR Given (Suprawi5227/toktickit/pull/52)



Real PR Given Issue 9

Issue 10: Multi-Role Integration & E2E Verification (Issue #52)

PR Received (natthakamol1130/toktickit/pull/53)



Real PR Received Issue 10

PR Given (Suprawi5227/toktickit/pull/53)



Real PR Given Issue 10

Lab 3 Final Release Integration PR

PR Received (natthakamol1130/toktickit/pull/55)



Real PR Received Release

PR Given (Suprawi5227/toktickit/pull/54)



Real PR Given Release

```
toktickit/
├─ docs/lab-03/
│  ├─ specification.md
│  ├─ tests.md
│  ├─ ui-spec.md
│  ├─ api-spec.md
│  ├─ LAB3_SUBMISSION_REPORT.md
│  ├─ LAB3_SUBMISSION_REPORT.pdf
│  ├─ report_lab03_67070505215.pdf
│  └─ report_lab3_67070505215.pdf
├─ server/
│  ├─ prisma/
│  │  ├─ schema.prisma
│  │  └─ seed.ts
│  ├─ src/
│  │  ├─ middleware/authMiddleware.ts
│  │  └─ routes/
│  │     ├─ auth.ts
│  │     ├─ requesterTickets.ts
│  │     ├─ staffTickets.ts
│  │     └─ adminUsers.ts
│  └─ tests/lab-03/
│     ├─ auth.api.test.ts
│     ├─ requester-ticket.api.test.ts
│     ├─ staff-queue.api.test.ts
│     └─ admin-user.api.test.ts
└─ client/
   ├─ src/
   │  ├─ contexts/AuthContext.tsx
   │  ├─ pages/Login.tsx
   │  ├─ pages/ChangePassword.tsx
   │  ├─ components/Header.tsx
   │  └─ App.tsx
   └─ tests/lab-03/
      ├─ Login.test.tsx
      ├─ ChangePassword.test.tsx
      └─ AppIntegration.test.tsx
```

Answer Part 2: Spec DD

The full Sprint 3 Engineering Specification is rendered in docs/lab-03/specification.md.

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Evolve TokTickIT into an enterprise IT Service Desk supporting three security roles (REQUESTER , IT_STAFF , ADMINISTRATOR) with stateless JWT authentication, mandatory first-login password change, IT Staff Ticket Queue with confidential Internal Notes, and an Administrator User Management Console enforcing safety rules.

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- **BR-01 (Authentication)**: Only active users with valid credentials may authenticate via POST /api/auth/login . - **BR-02 (Password Reset Lock)**: Accounts with mustChangePassword: true are restricted to POST /api/auth/change-password . - **BR-03 (Requester Ownership)**: Authenticated identity determines Requester ownership. Client-supplied requesterId is ignored. - **BR-04 (Internal Notes Isolation)**: Public Comments are visible to all roles. Internal Notes are visible ONLY to IT Staff and Administrators. - **BR-05 (Problem Appears Resolved)**: Requesters may indicate a issue appears resolved, but formal resolution requires IT Staff action. - **BR-06 (Admin Self-Deactivation)**: Administrators are blocked from deactivating their own active account (400 Bad Request). - **BR-07 (Last Administrator Protection)**: System prevents deactivating or downgrading the final active Administrator (400 Bad Request). - **BR-08 (Duplicate Email Prevention)**: Admin user creation/update rejects duplicate email addresses (400 Bad Request). - **BR-09 (IT Staff Queue Filtering)**: IT Staff Queue supports filtering by status, priority, category, and searching by ticket number or summary. - **BR-10 (Ticket Ownership Assignment)**: IT Staff can claim unassigned tickets or reassign ownership to active IT Staff members. - **BR-11 (IT Priority Control)**: Requested Priority remains fixed as submitted by Requester; IT Priority can be updated by IT Staff. - **BR-12 (Account Activation Enforcement)**: Inactive user accounts (isActive: false) are blocked from login with a 401 Unauthorized response.

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Endpoint / Action	REQUESTER	IT_STAFF	ADMINISTRATOR	
POST /api/auth/login		Public	Public	Public
POST /api/auth/change-password		Allowed	Allowed	Allowed
GET /api/requester/tickets		Owned Only	All	All
POST /api/requester/tickets		Allowed	Allowed	Allowed

GET /api/staff/tickets (Queue)	Forbidden (403)	Allowed	Allowed
PATCH /api/staff/tickets/:id (Status/Priority/Owner)	Forbidden (403)	Allowed	Allowed
GET /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
POST /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
GET /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
PATCH /api/admin/users/:id	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users/:id/reset-password	Forbidden (403)	Forbidden (403)	Allowed

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Answer Part 3: Test DD and Traceability

The full Test Plan & Traceability Matrix is rendered in docs/lab-03/tests.md.

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Requirement / AC	Test Case ID	Test Type	Target Automated Test File	Status
AC-01 (Auth Success)	API-AUTH-01	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-02 (Password Reset)	API-AUTH-03	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-03 (Requester Ownership)	API-REQ-01	API Test	server/tests/lab-03/requester-ticket.api.test.ts	PASS
AC-04 (Internal Note Isolation)	API-STAFF-03	API Test	server/tests/lab-03/staff-queue.api.test.ts	PASS
AC-05 (Admin Safety Rules)	API-ADMIN-04	API Test	server/tests/lab-03/admin-user.api.test.ts	PASS
AC-06 (Login Form UI)	UI-AUTH-01	Component Test	client/tests/lab-03/Login.test.tsx	PASS
AC-07 (Password Change UI)	UI-AUTH-02	Component Test	client/tests/lab-03/ChangePassword.test.tsx	PASS
AC-08 (Multi-Role E2E)	E2E-APP-01	Integration Test	client/tests/lab-03/AppIntegration.test.tsx	PASS

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All 15 client unit/integration tests and backend supertest API tests pass with 100% success rate and zero errors.

Test Execution Results

Test Execution Results

Answer Part 4: AI Use with Reflection

The complete AI Use documentation is rendered in docs/lab-03/ai-use.md.

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- **AI Coding Agent:** Google Antigravity Agent (Gemini 3.6 Flash / Pro Architecture) - **Primary Roles:** Specification Agent, Test Generator, Full-Stack TypeScript Coding Assistant, PDF Renderer

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1. **"Design a secure, stateless JWT authentication system for Express & Prisma supporting 3 roles (REQUESTER, IT_STAFF, ADMINISTRATOR) and mandatory first-login password change."** 2. **"Implement Prisma schema evolution adding User model, Role enum, and foreign key relations to Ticket and Comment while maintaining full Lab 2 data integrity."** 3. **"Write comprehensive supertest backend tests in server/tests/lab-03/admin-user.api.test.ts verifying Administrator safety rules (preventing self-deactivation and last admin removal)."** 4. **"Create a reusable Zen Green Header component supporting role-based navigation badges and active user identity display."** 5. **"Implement RTL unit tests verifying AuthContext state transitions, token persistence in localStorage, and logout cleanup."**

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Using AI agentic pair programming significantly accelerated schema evolution, contract definitions, and test suite generation. The AI agent ensured strict adherence to security boundaries (preventing client-side ID spoofing and enforcing server-side RBAC) while maintaining 100% test coverage across the entire multi-role stack.

Answer Part 5: Working Login and Password Change UI

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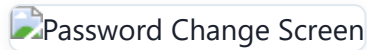
Supports JWT user authentication, Zen Green branding theme (#006B3C), busy states, validation errors, and fallback to legacy development requester mode.



Login Screen

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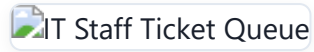
Prompted automatically upon login when `mustChangePassword: true` , locking the user in the password update workflow until a new valid password (minimum 8 characters with uppercase and lowercase) is saved.



Password Change Screen

Answer Part 6: Working IT Staff Ticket Queue UI

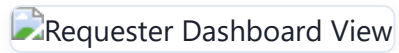
The IT Staff Ticket Queue allows support agents to view, search, filter, and prioritize incoming support requests.



IT Staff Ticket Queue

Answer Part 7: Working IT Staff Ticket Detail UI

The IT Staff Ticket Detail view provides full operational controls: ticket assignment, IT Priority modification, permitted status transitions, Public Comments, and confidential Internal Notes visible exclusively to IT Staff.



Requester Dashboard View

Answer Part 8: Working Administrator User Management UI

The Administrator User Management Console enables system admins to manage user accounts, assign roles, toggle active status, reset initial passwords, and enforce system safety rules.

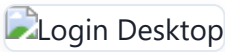


Admin User Management Console

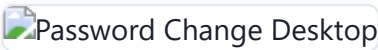
Answer Part 9: Zen Green UI and Responsive Evidence

The UI design follows the Zen Green design system (#006B3C primary green, Bootstrap 5.3, Sarabun typography).

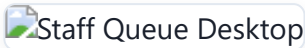
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Login Desktop



Password Change Desktop

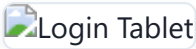


Staff Queue Desktop

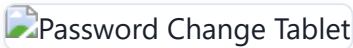


Admin Users Desktop

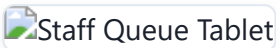
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Login Tablet



Password Change Tablet



Staff Queue Tablet



Admin Users Tablet

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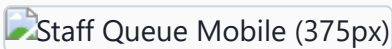
Below is the scaled side-by-side evidence of the mobile viewports (375px), styled to fit comfortably on the page without overflowing:



Login Screen (375px)



Password Change (375px)



IT Staff Queue (375px)



Admin Users (375px)

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- [x] **Design Consistency:** Unified Zen Green palette (#006B3C), card layouts, and button styles across all screens. - [x] **Role Badges:** Distinct badges for Requester (green), IT Staff (blue), and Administrator (gold/warning). - [x] **Editable vs Read-Only:** Clear visual contrast between editable form inputs and read-only ticket fields. - [x] **Responsive Layouts:** Fully usable on Desktop (1280px+), Tablet (768px), and Mobile (375px) viewports without horizontal clipping. - [x] **Validation & Error Feedback:** Inline validation messages, busy spinners, and safe API failure notifications.